FEDERAL SPECIFICATION

GILSONITE, NATURAL ASPHALT (PAVEMENT SEALER)

This specification was approved by the Commissioner, Federal Supply Services, General Services Administration for use of all Federal agencies.

1. SCOPE

1.1 Scope. This specification covers one type of natural asphalt (Gilsonite) pavement sealer used as treatment on asphalt concrete pavements.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-D-729 - Drums, Metal, 55 Gallon (For Shipment of Noncorrosive Materials).
PPP-P-704 - Pails, Metal: Shipping, Steel (1 through 12 Gallon).

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

D5 - Test for Penetration of Bituminous Materials.
D36 - Test for Softening Point of Asphalts and Tar Pitches (Ring and Ball Apparatus).
D95 - Test for Water in Petroleum and Other Bituminous Materials.

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D140 - Sampling Bituminous Materials.
D402 - Test for Distillation of Cut-Back Asphaltic Products.
D1074 - Test for Compressive Strength of Bituminous Mixtures.
D1559 - Test for Resistance to Plastic Flow of Bituminous Mixtures Using Marshall Apparatus.
D2042 - Test for Solubility of Bituminous Materials in Organic Solvents.
D2170 - Test for Kinematic Viscosity of Asphalts.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

Technical society and technical association specifications and standards are generally available for references from libaries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Description. The compound shall consist of Gilsonite and light solvent. Other ingredients may be added providing the compound conforms to the requirements stated herein.

3.2 Working properties. The compound shall be of a suitable consistency for uniform application with brush or airless spray at ambient temperatures above 32°F(0°C) and at a covering rate of approximately 0.1 gallon per square yard (see 6.1).

3.3 Condition in container. The compound shall be a smooth, homogenous mixture, ready to apply without heating, thinning or stirring.

3.4 Stickiness or tackiness. When tested in accordance with 4.3.2, the surface of the specimens shall not feel sticky or tacky to the touch after 2 hours of curing.

3.5 Stripping loss. When tested in accordance with 4.3.3, the material stripped from the sample shall not exceed 10 percent by weight.

3.6 HCL precipitation value. When tested in accordance with 4.3.4, the HCL precipitation value shall be not less than 30.

3.7 Property requirements. The sealer shall conform to the requirements of Table I when tested as specified in Table II.

TABLE I. Requirements

Characteristic	Minimum	Maximum
Kinematic viscosity at 140°F, cs.	10	20
Residue from distillation to 680°F, percent volume	30	—
Residue from distillation		
Penetration at 77°F, 5 sec., 100 gms	2	5
Softening point, deg. F	190	
Solubility in CCL4, percent	99	
Water, percent		0.1

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 Lot. Unless otherwise specified (see 6.2), a lot shall consist of all of the pavement sealer offered for delivery at the same time.

4.2.2 Sampling for tests. Sampling for tests shall be in accordance with MIL-STD-105, using the sampling methods specified in ASTM D140. Samples shall be stored in clean, airtight, sealed containers at a temperature of not less than 40°F, until tested.

4.3 Test methods.

4.3.1 The sealer shall be tested in accordance with the methods indicated in Table II, and the test methods described in 4.3.2, 4.3.3 and 4.3.4. Failure to pass any test and non-compliance with the requirements shall be the basis for rejection of the lot. The samples selected in accordance with 4.2.1 shall be used in performing all tests.

TABLE II. Test Methods

Properties	Applicable Tests
Kinematic viscosity	ASTM D2170
Distillation	ASTM D402
Tests on residue from distillation	
Penetration	ASTM D5
Softening point	ASTM D36
Solubility in CCL4	ASTM D2042
Water	ASTM D95

4.3.2 Stickiness or tackiness. Prepare specimen by mixing 90 parts graded Ottawa sand and 10 parts Portland cement with 5 parts 85-100 penetration or AC20 Grade asphalt. Compact mixture in accordance with either ASTM D1074 or ASTM D1559. Spray specimen with sealer at the rate of 0.1 gallon per square yard, place in oven at 110°F(43.3°C) for 2 hours, remove and cool for not more than 10 minutes, and touch lightly with fingers.

4.3.3 Stripping loss.

(a) The aggregate used shall be meta-andesite, unwashed and oven dried at 230°F(110°C), passing a 3/8 inch sieve and retained on a No. 4 sieve.
(b) Mix 150 grams (Weigh to Nearest 0.1 gram) of aggregate and 15 grams of sealer until the aggregate is completely coated.
(c) Pour mixture onto a 6 inch square No. 20 mesh screen drain. Cure in oven 24 hours at 140°F(50°C).
(d) Remove sample from screen, weigh, and place sample in a 1 gallon metal can having a compression lid.
(e) Fill the can with cool tap water (pH 6.0 to 8.0) and allow it to stand at room temperature for 24 hours.
(f) Seal the can and agitate on a paint can shaker for 10 minutes.
(g) Decant and drain sample, dry with absorbent paper, and place in a dessicator at room temperature for 24 hours.
(h) Weigh sample and calculate the percentage stripped from sample.

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4.3.4 HCL precipitation value. Weigh a 1.0 gram sample, to the nearest milligram (0.001 gram), of the residue obtained from the ASTM D402 distillation into a tared centrifuge tube. Add 50 ml benzene and shake until solids dissolve. Add 1.0 ml concentrated hydrochloric acid, stopper and shake vigorously for two minutes. Allow to stand for 15 minutes, then centrifuge at 1500 RPM to 2000 RPM for 10 minutes. Decant benzene, add 50 ml benzene and re-slurry the precipitate by means of a thin wire and shake tube for 1 or 2 minutes. Centrifuge for 10 minutes and decant benzene. Repeat this wash procedure one additional time. After decanting final spin, place tube in 110°C oven until constant weight is achieved (about 4 hours). Place in dessicator 1 hour, then weigh. Calculate the HCL precipitate value as the precipitate percentage of the sample.

4.4 Examination of preparation for delivery. An examination shall be made during the preparation for delivery to make certain that the packing and marking of the containers conform to the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be Level A, B, or C as specified (see 6.2).

5.1.1 Levels A and B. The pavement sealer shall be packed in 55 gallon drums conforming to PPP-D-729, Type II, or in 5-gallon pails conforming to PPP-P-704, Type I class 1, 2, 3, 4, 5, or 8.

5.1.2 Level C. The pavement sealer shall be packed in quantities as specified (see 6.2) in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.2 Marking.

5.2.1 Civil Agencies. In addition to markings required by the contractor or order, the pails and drums shall be marked in accordance with Fed. Std. 123.

5.2.2 Military Activities. In addition to markings required by the contractor or order, the pails and drums shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The pavement sealer is intended for use in coating, crack filling and treating asphalt concrete. For optimum results apply with ambient temperature above 55°F(12.8°C).

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Container size (see 5.1.1).
(c) Size of lot, if different from 4.2.1.
(d) Selection of level of packing required (see 5.1).

Preparing activity:

GSA-FSS

Civil Agency Coordinating Activities:

COMMERCE-NBS

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